



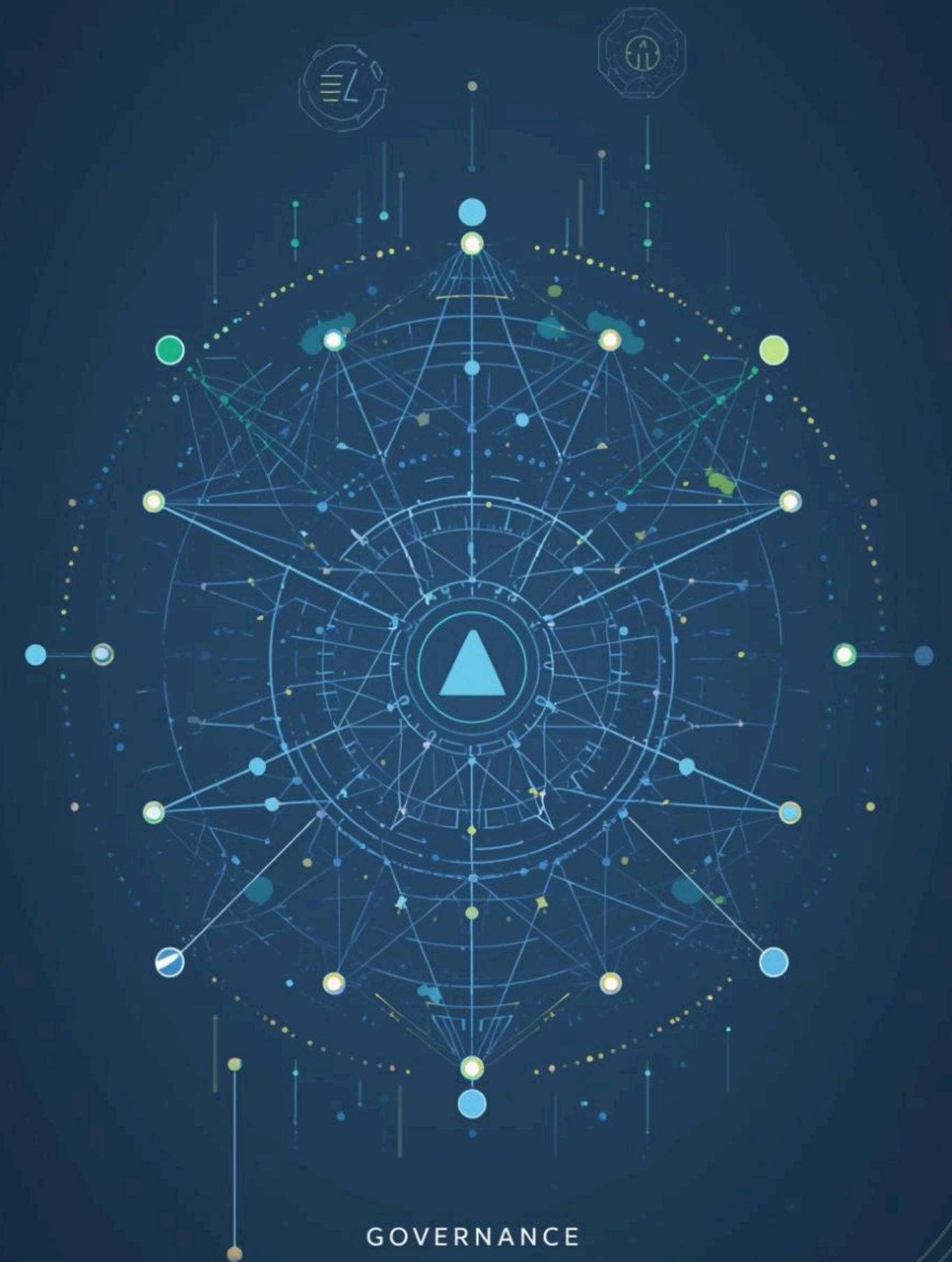
Regulatory Approaches



Why?

Markets fail with AI:

- Externalities - Companies don't bear full costs of harms
- Information asymmetry - Users can't assess risks
- Power imbalances - Individuals can't negotiate with platforms
- Race to bottom - Competition undermines safety



Four Regulatory Models

- Self-Regulation
- Sectoral Regulation
- Horizontal Regulation (EU AI Act)
- Adaptive Regulation



Self Regulation

Industry develops own standards, ethics boards, best practices

- Pros:
 - Flexible, adapts quickly
 - Industry expertise
 - Less bureaucratic burden
 - Innovation-friendly
- Cons
 - Fox guarding henhouse
 - Minimal enforcement
 - PR over substance
 - Race to bottom (companies that ignore standards gain advantage)



Sectoral Regulation

Existing regulators address AI in their domains e.g FDA for medical AI

- Pros:
 - Domain expertise
 - Uses existing regulatory infrastructure
 - Targeted to specific risks
- Cons
 - Fragmented approach
 - Regulators may lack AI expertise
 - Gaps where AI doesn't fit existing categories
 - Slow to adapt to new AI capabilities



Horizontal Regulation

Comprehensive AI specific law with strict rules for high-risk and minimal for low-risk

- Pros:
 - Comprehensive
 - Technology-specific expertise can develop
 - Clear rules across sectors
- Cons
 - Risk of overregulation stifling innovation
 - Difficult to keep pace with technology
 - May be too broad or too narrow
 - Enforcement challenges



Adaptive Regulation

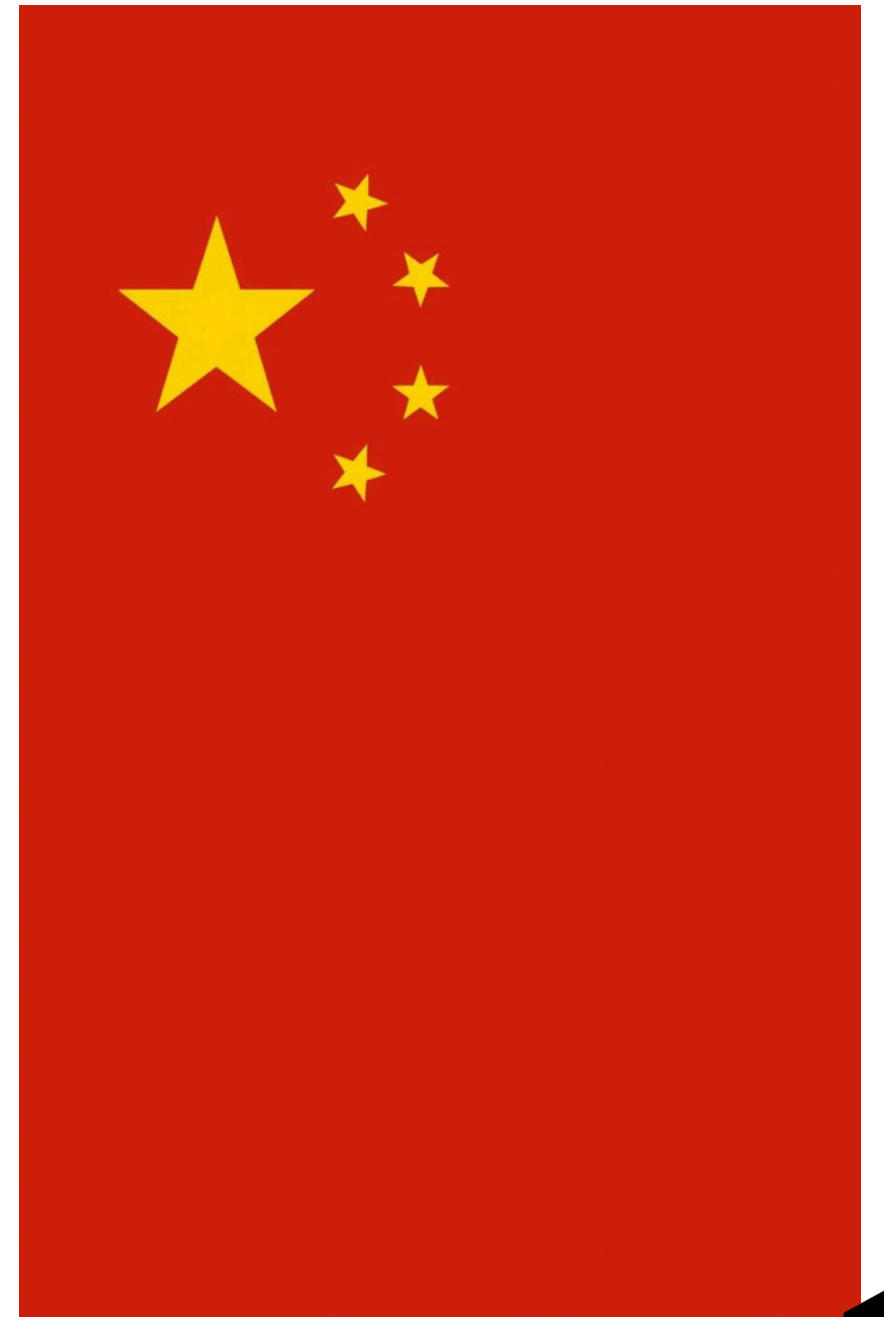
Regulatory sandboxes with continuous monitoring

- Pros:
 - Balances innovation and safety
 - Learns from real-world deployment
 - Adapts to technological change
- Cons
 - Resource-intensive
 - Unequal access (benefits large companies with resources to participate)



Geographic Comparison

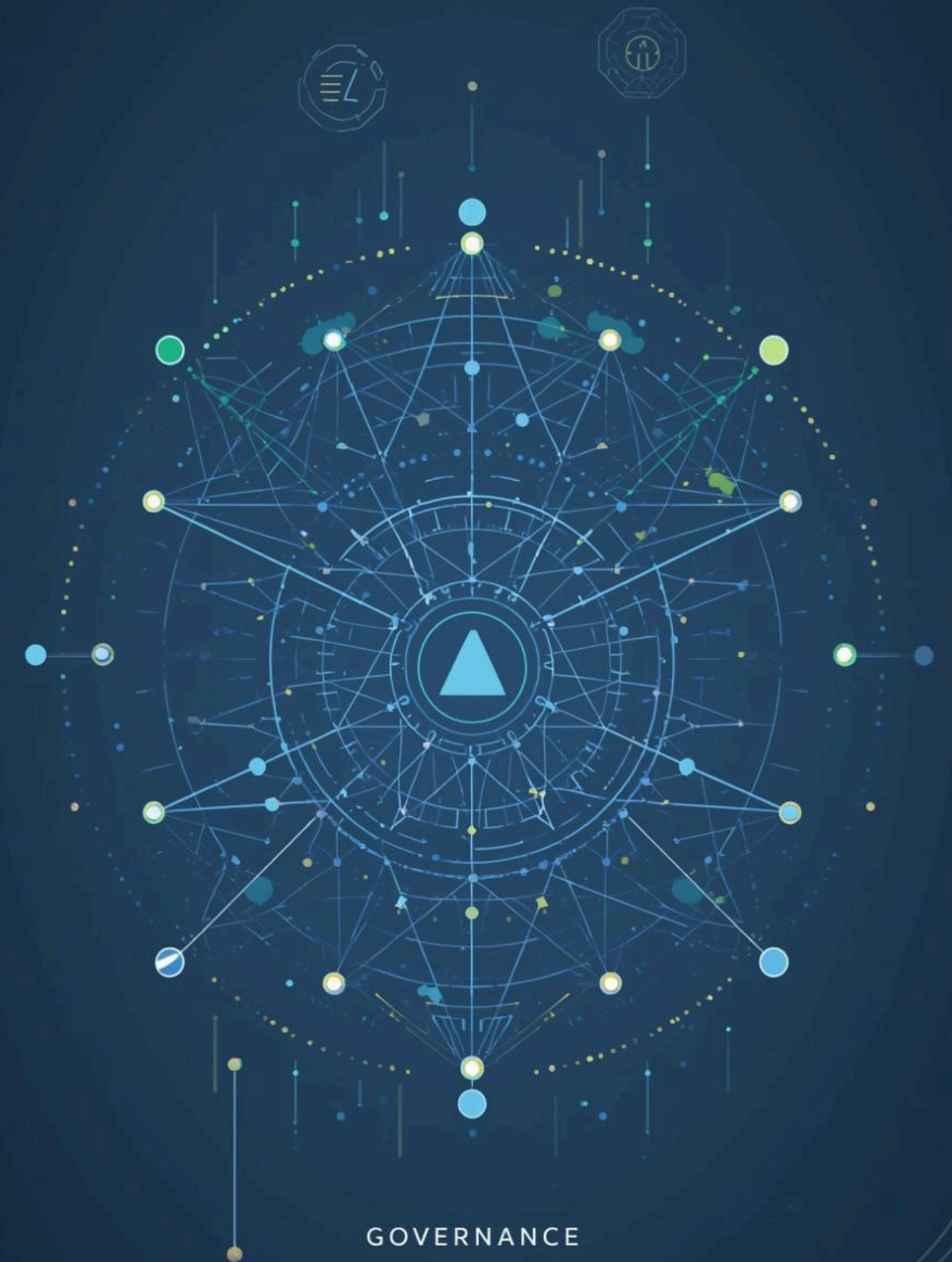
- US: Light-touch, sectoral, innovation first
- EU: Comprehensive regulation, precautionary principle
- China: State-directed, focused on stability and control



Challenges

Algorithmic transparency: Should companies explain decisions?

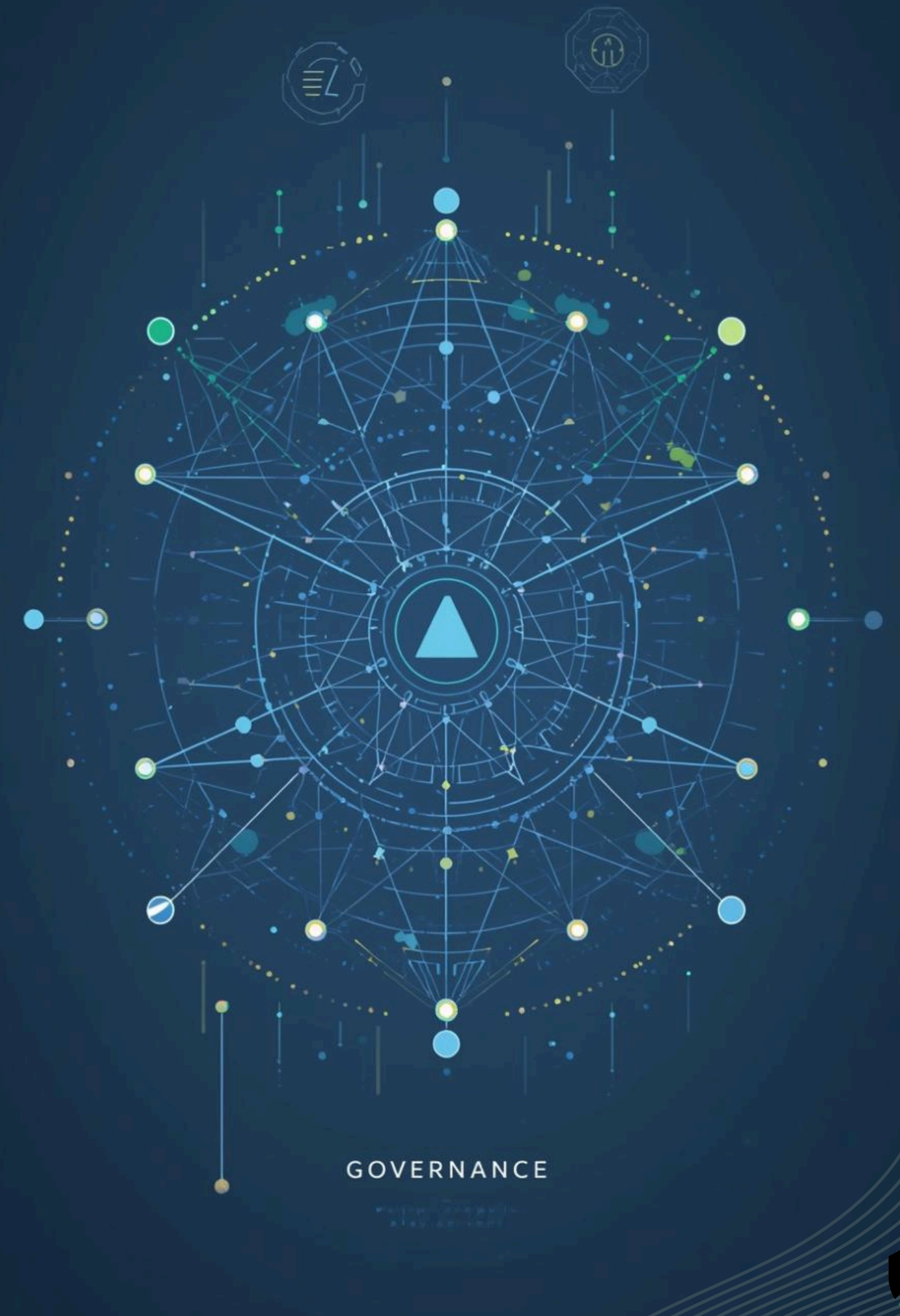
- For: Accountability, identify bias, democratic oversight
- Against: Trade secrets, technical impossibility, gaming risk



Challenges

Liability Framework: Who pays when AI causes harm?

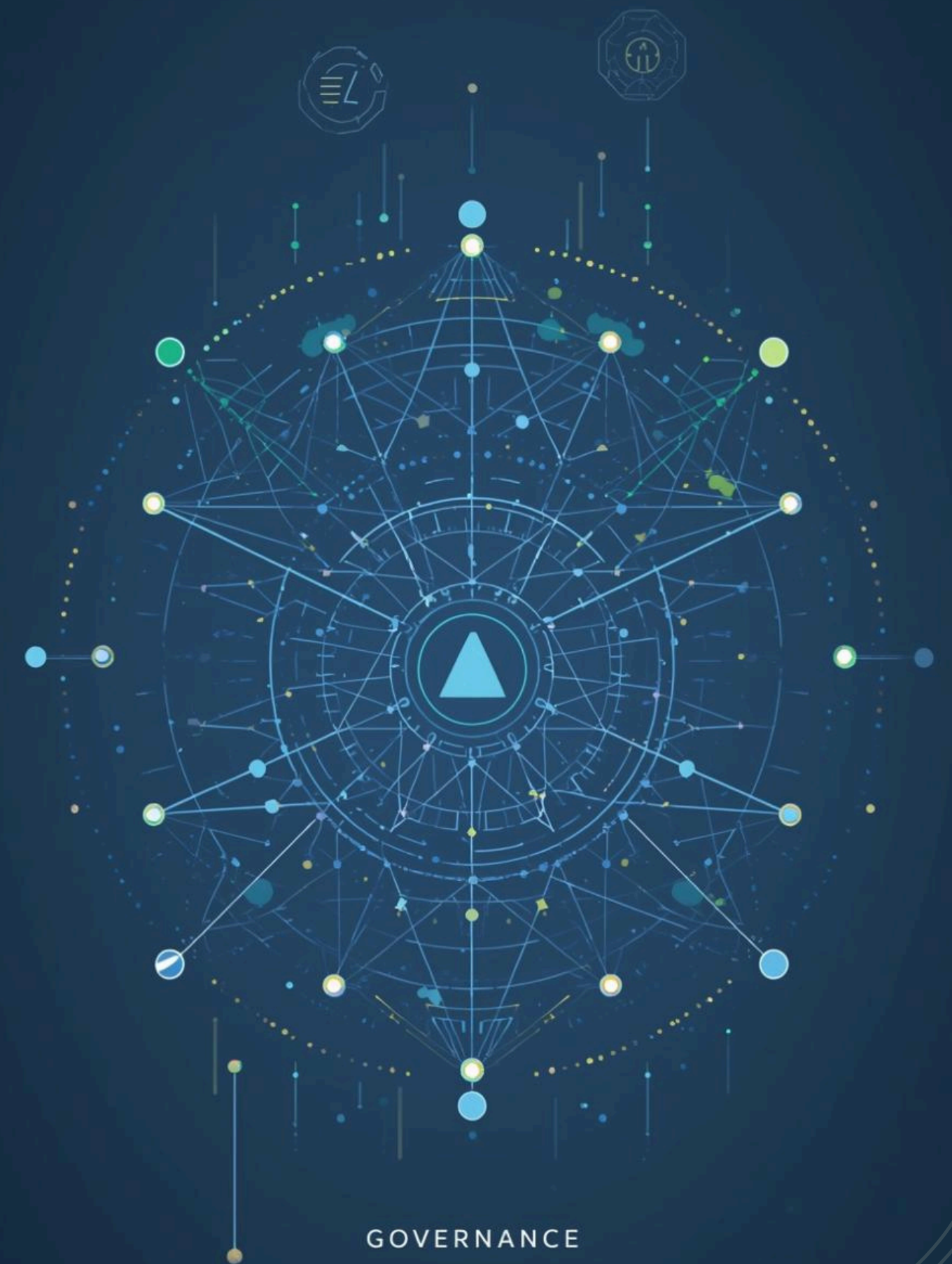
- Developer liability (encourages safety, but may slow innovation)
- User/deployer liability (those using AI responsible)
- Shared liability (apportioned based on control/foreseeability)
- Strict liability (no need to prove negligence)
- No-fault insurance schemes



Challenges

Pre-deployment testing: Like FDA drug approval but far more challenging

- Infinite possible inputs
- Deployment context matters
- Emergent behaviors
- Continuous learning systems



Challenges

Auditing and Certification: Third-party audits of AI systems?

- Potential benefits:
 - Independent verification
 - Standardized assessments
 - Market for safety/ethics expertise
- Challenges:
 - Who audits? (Conflict of interest)
 - What standards? (Metrics for fairness, safety)
 - Access to proprietary systems
 - Continuous vs. one-time audits

